

SBT-DF202

Lab 1: Digital Forensics Case Handling,

Autopsy and Sleuth Kit Analysis

Okunlola Emmanuel Oluwasegun

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LAB OVERVIEW

This laboratory exercise simulates a structured digital forensic investigation within the International Center for Digital Forensic and Cyber Crime Analysis (ICDFA) framework. Acting as a junior digital forensic analyst, the practitioner is tasked with conducting a comprehensive, non-destructive examination of a raw forensic disk image designated as Ch01InChap01.dd. The investigation encompasses pre-analysis evidence preservation, graphical user interface (GUI) analysis using Autopsy, low-level command-line interface (CLI) structural analysis using The Sleuth Kit (TSK), deleted file recovery, metadata evaluation, keyword searching, and cryptographic hash verification. All investigative steps must adhere strictly to forensic principles, ensuring that the original evidence remains untampered and that all actions are fully documented, repeatable, and preserved within a structured audit trail.

LEARNING OBJECTIVES

The objectives of this lab are to make the students:

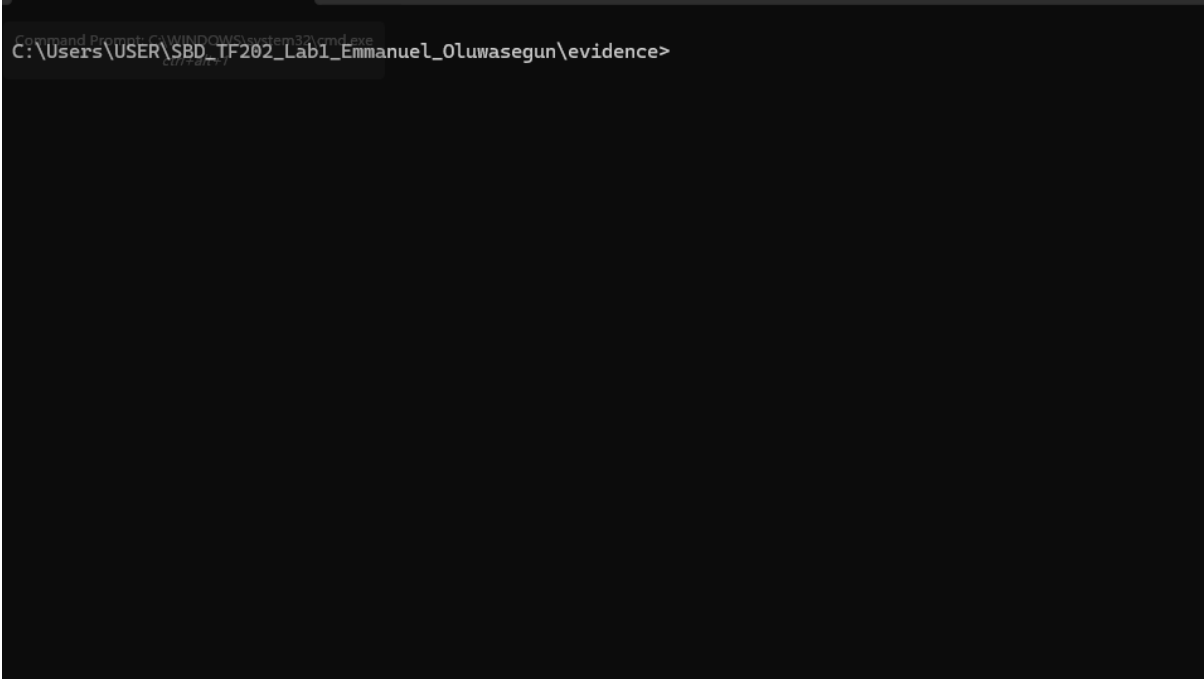
1. Preserve and Authenticate Digital Evidence.
2. Execute GUI-Based Forensic Analysis.
3. Conduct Command-Line Disk Structure Analysis.
4. Examine Metadata and File System Inodes.
5. Perform Targeted File Recovery.
6. Validate Recovered Evidence via Hash Comparison.

MATERIALS and TOOLS

1. Autopsy
2. The Sleuth Kit (TSK)
3. CertUtil
4. md5sum & sha256sum
5. Microsoft Word & GitHub.

METHODOLOGY

The authorized training forensic image Ch01InChap01.dd was downloaded and then a forensic working directory was created before carrying out the analysis.



```
Command Prompt: C:\WINDOWS\system32\cmd.exe
C:\Users\USER\SBD_TF202_Lab1_Emanuel_Oluwasegun\evidence>
```

The image above shows the working directory that was created before starting the analysis.

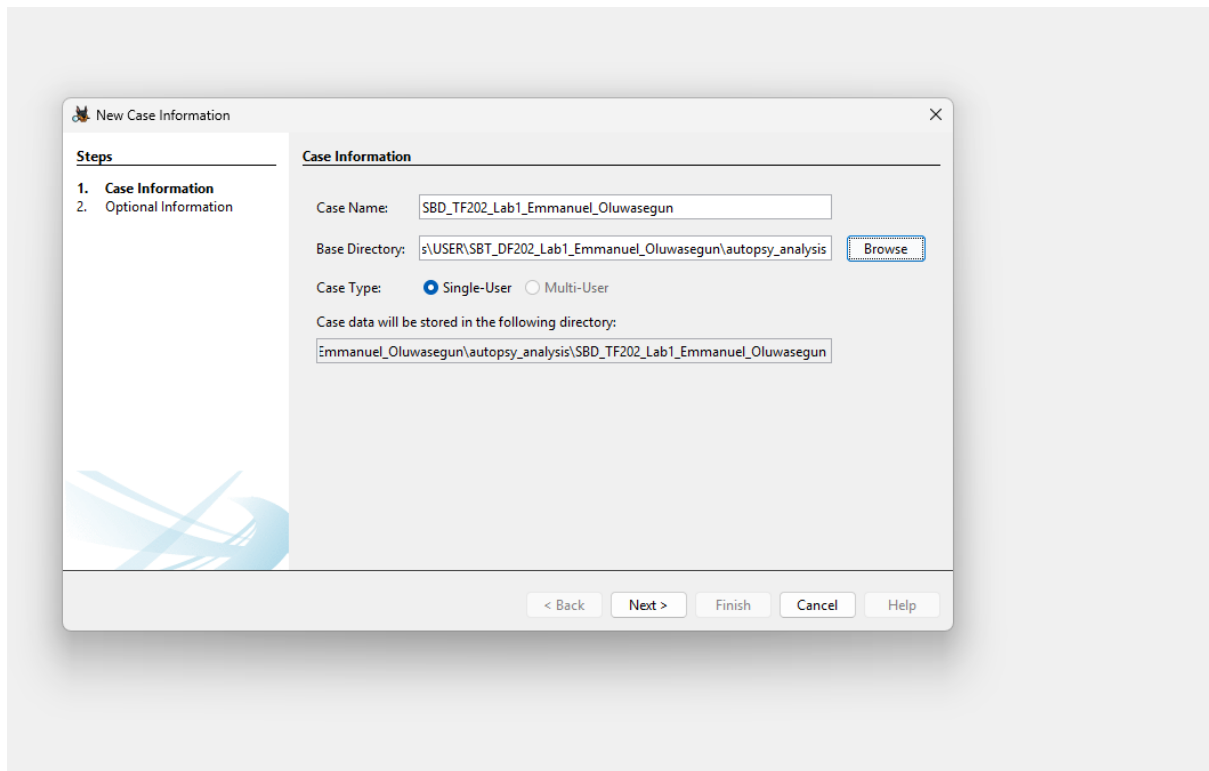
PRE-ANALYSIS VERIFICATION TABLE

Evidence source	Ch01InChap01.dd (Downloaded from Authorized IC DFA Course Portal)
File size	1,474,560 bytes
Storage location	C:\Users\USER\SBT_DF202_Lab1_Emanuel_Oluwasegun\evidence\Ch01InChap01.dd
MD5 Hash	a117773bcf1fc88ec0ab8e0a349fbbcb
SHA 256 Hash	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122

AUTOPSY ANALYSIS

The first process involves using the Autopsy software- a GUI based program, which allowed me to efficiently analyze the forensic image (Ch01InChap01.dd).

The whole process and screenshots are explained below:



The first image above shows the creation of the case where the case name (SBT_DF202_Lab1_Emmanuel_Oluwasegun) and the base directory were inputted.

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: SBT-DF202-LAB1

Examiner

Name: Emmanuel Oluwasegun

Phone: 09026302193

Email: ematogs001@gmail.com

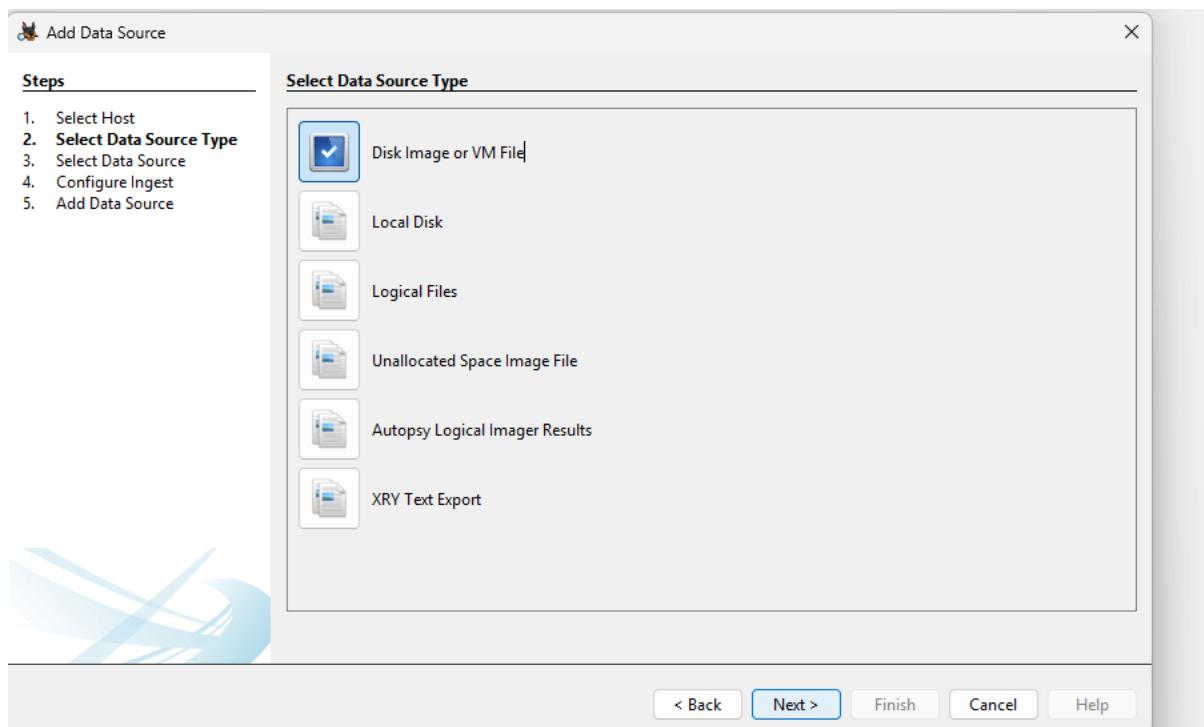
Notes: This is the very first case.

Organization

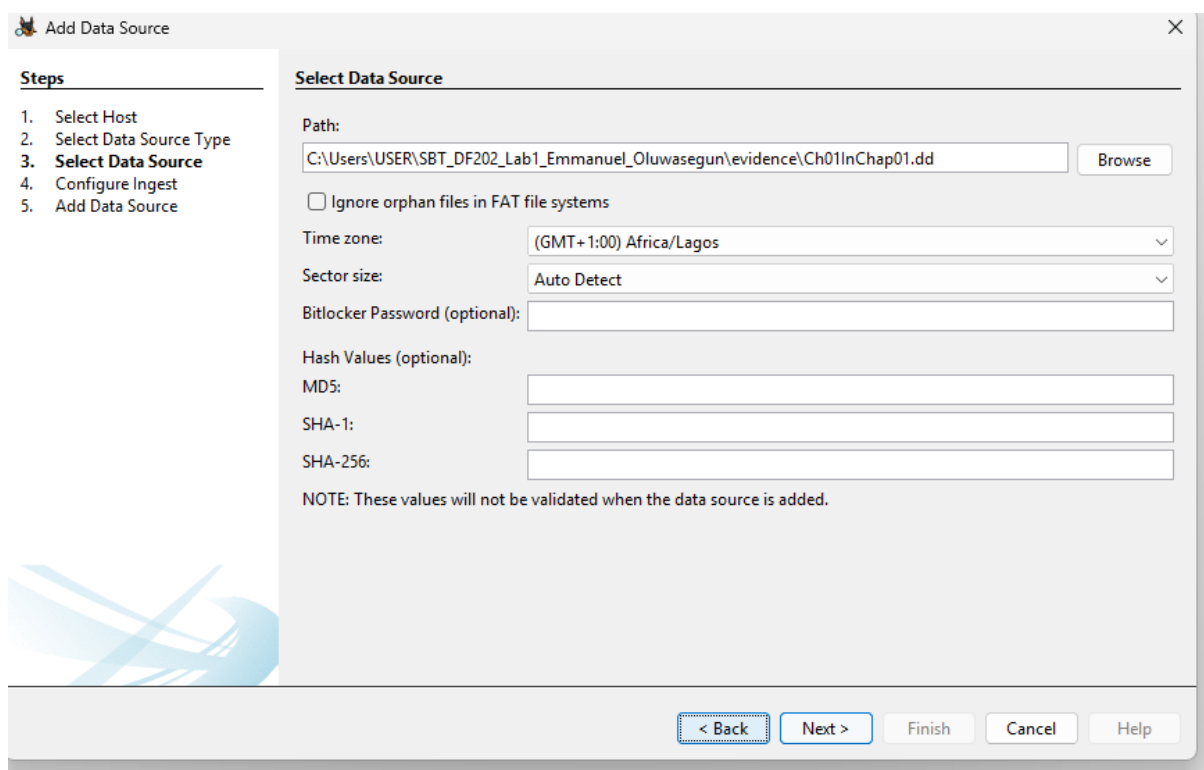
Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

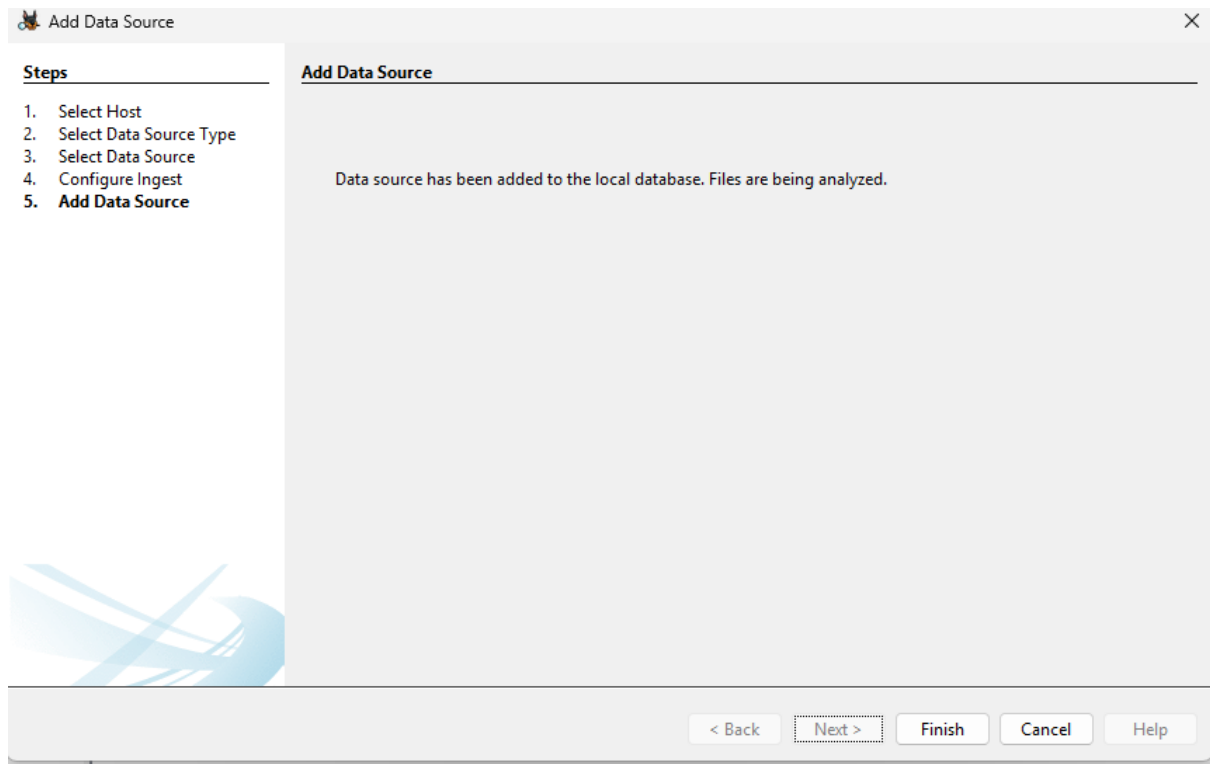
This image above shows the case information including the details of the Examiner.



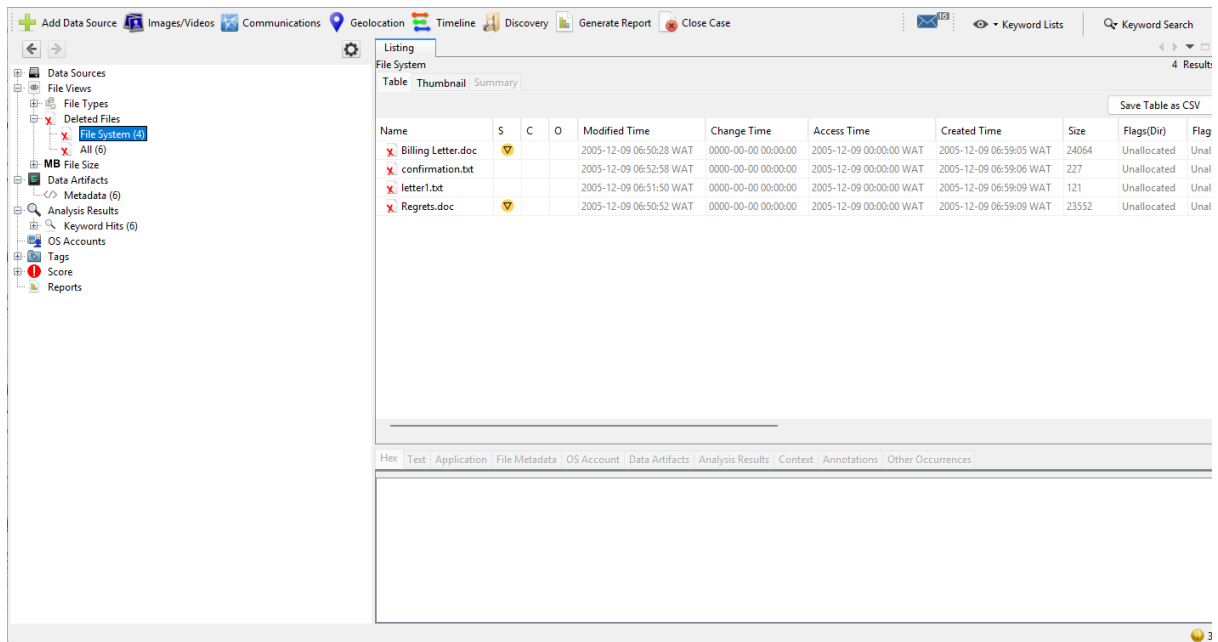
Here in this image, we selected the data source type.



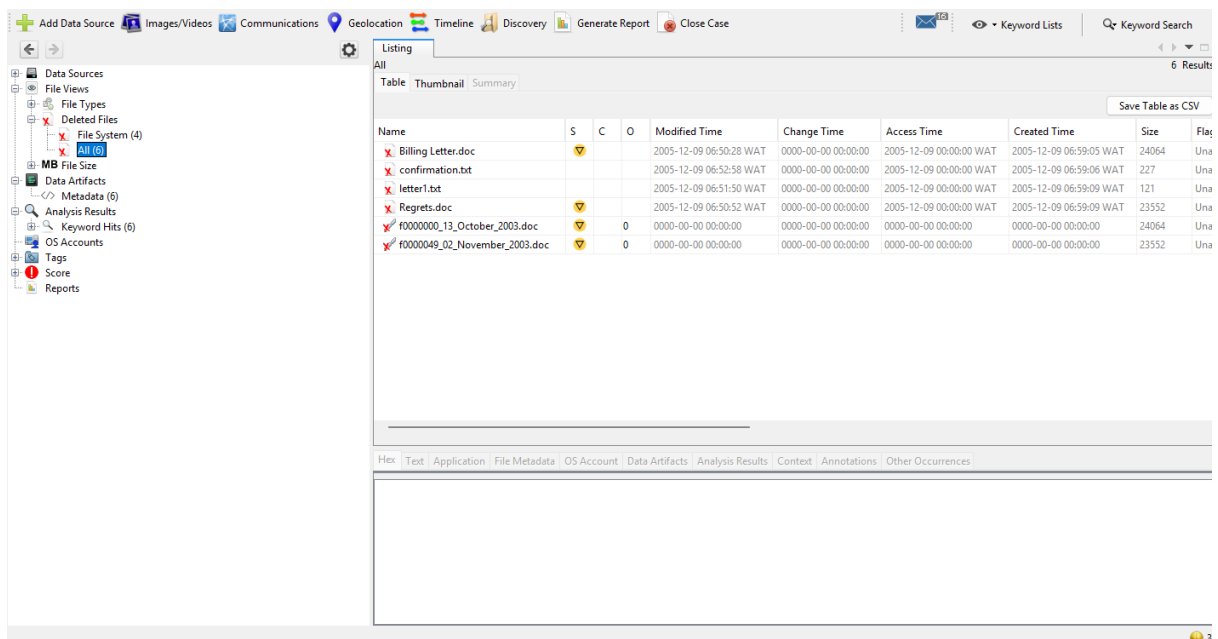
Here, we select the data source.



This image shows the end of the data source addition.



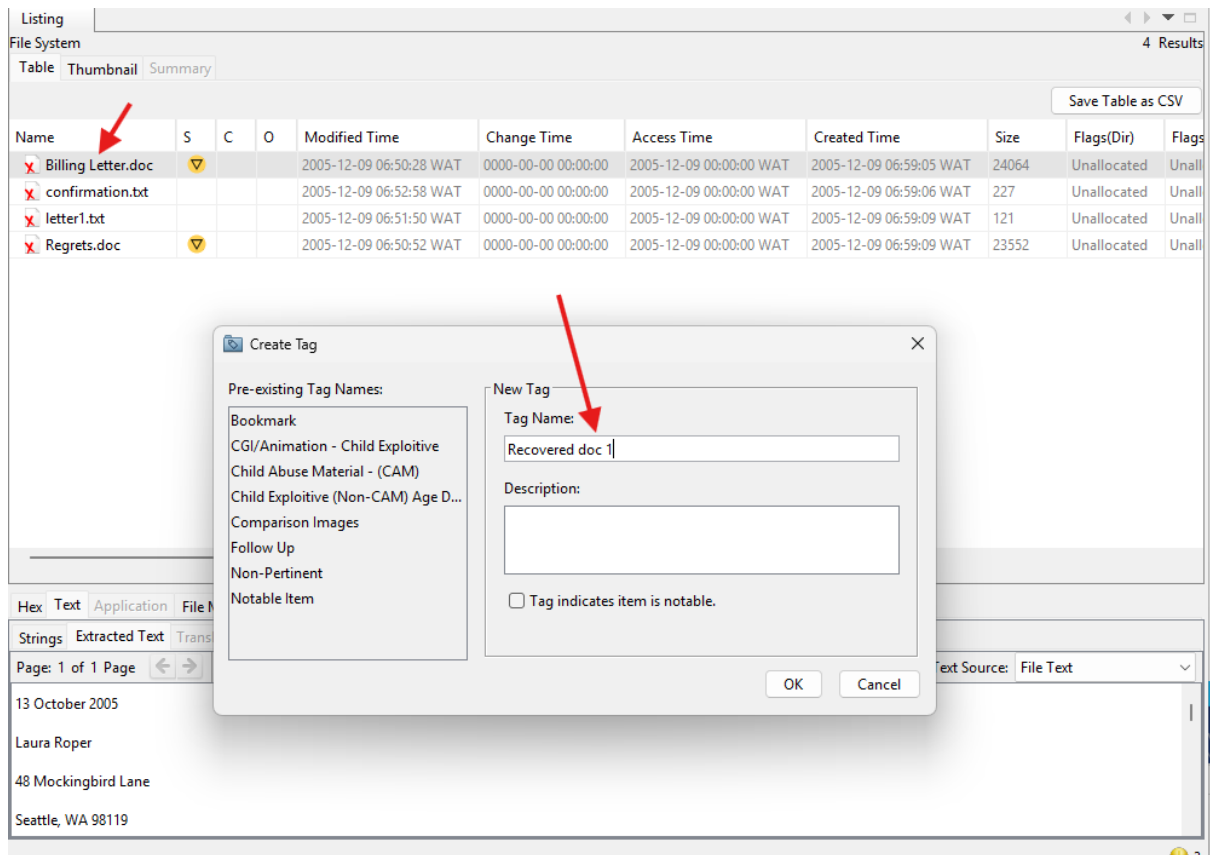
The image above confirms that Autopsy has finished ingesting the image file.



Here we have the deleted files entry.

TAGGING AND RECOVERY OF DELETED FILES

The images below explain the process of tagging the deleted files and then the recovery of the files.



The image above shows the tagging of the deleted files.

Listing

File System 4 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags
 Billing Letter.doc				2005-12-09 06:50:28 WAT	0000-00-00 00:00:00	2005-12-09 00:00:00 WAT	2005-12-09 06:59:05 WAT	24064	Unallocated	Unall
 confirmation.txt				2005-12-09 06:52:58 WAT	0000-00-00 00:00:00	2005-12-09 00:00:00 WAT	2005-12-09 06:59:06 WAT	227	Unallocated	Unall
 letter1.txt				2005-12-09 06:51:50 WAT	0000-00-00 00:00:00	2005-12-09 00:00:00 WAT	2005-12-09 06:59:09 WAT	121	Unallocated	Unall
 Regrets.doc				2005-12-09 06:50:52 WAT	0000-00-00 00:00:00	2005-12-09 00:00:00 WAT	2005-12-09 06:59:09 WAT	23552	Unallocated	Unall

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

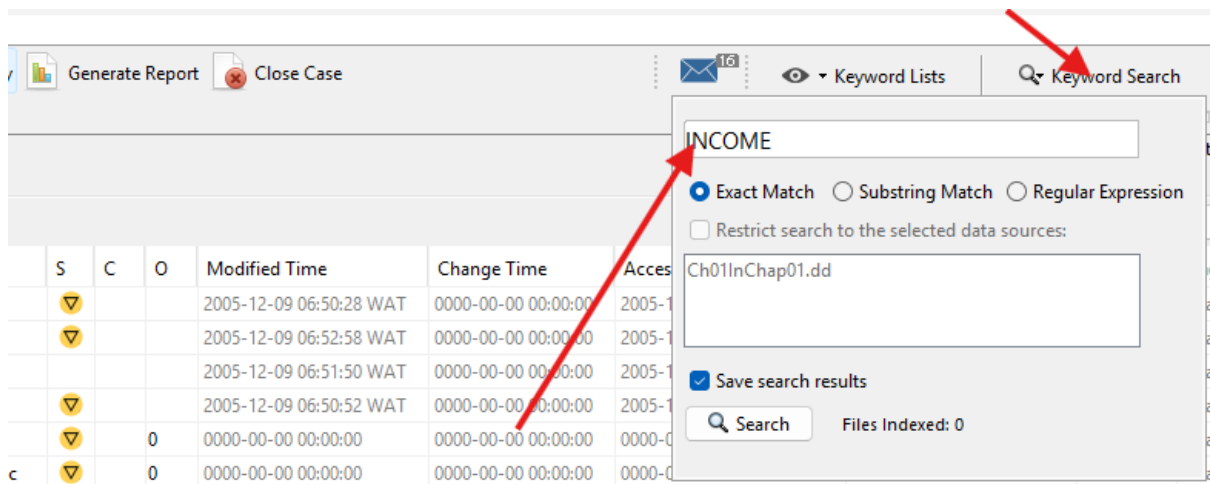
13 October 2005

Laura Roper

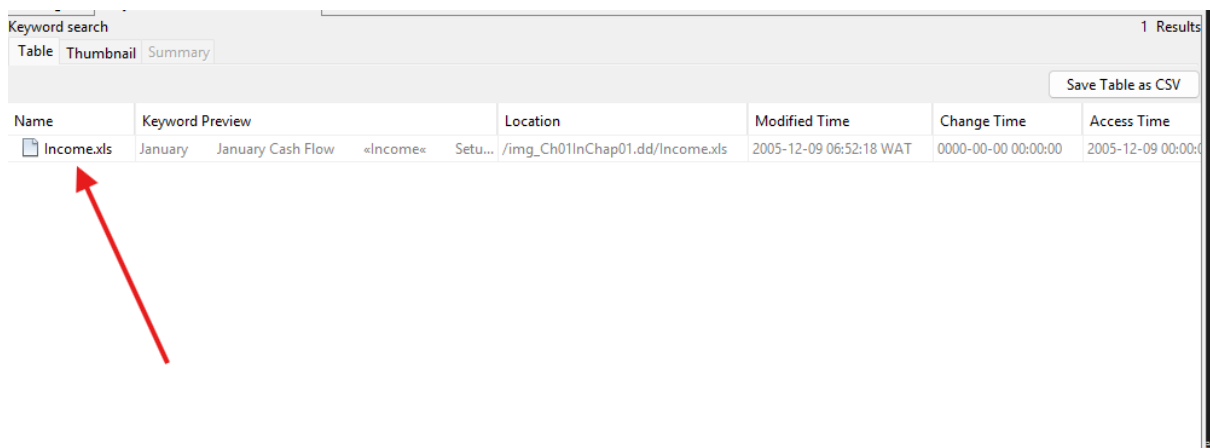
48 Mockingbird Lane

Seattle, WA 98119

This one above also depicts more deleted files already tagged.



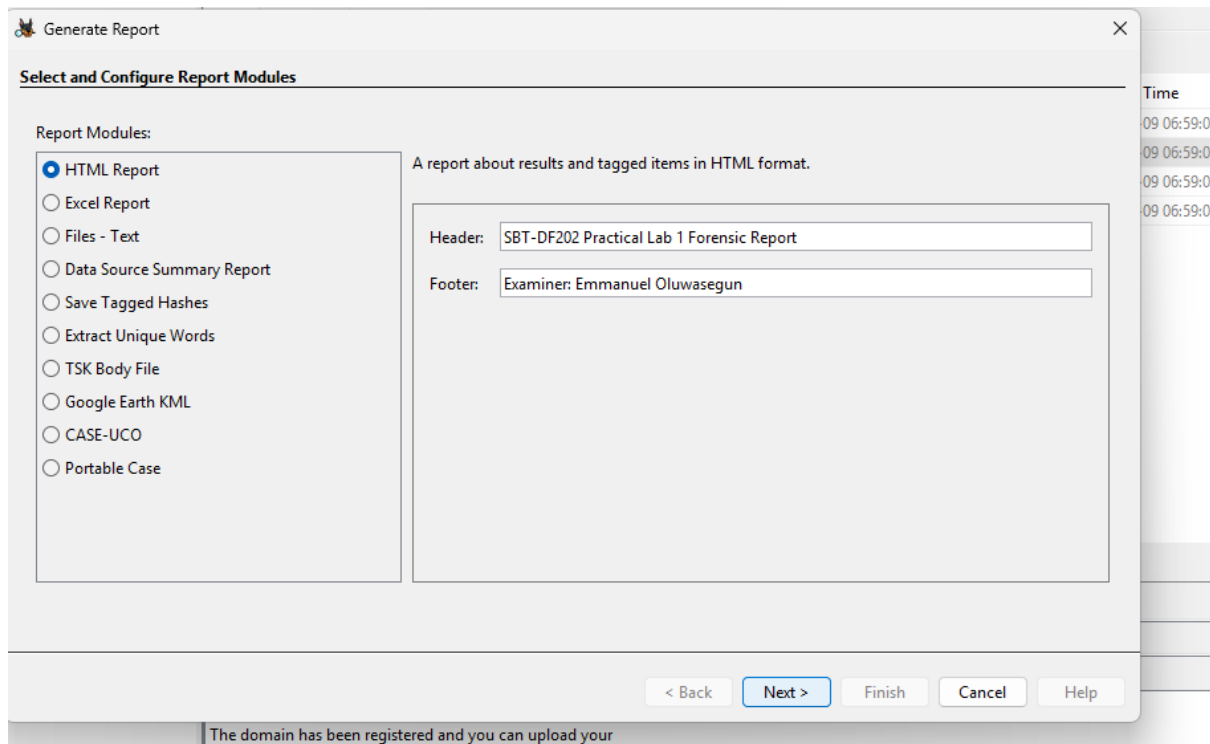
The image above shows the application 'keyword search' to see some hidden deleted files, we apply keyword search and then we can see the hidden files.



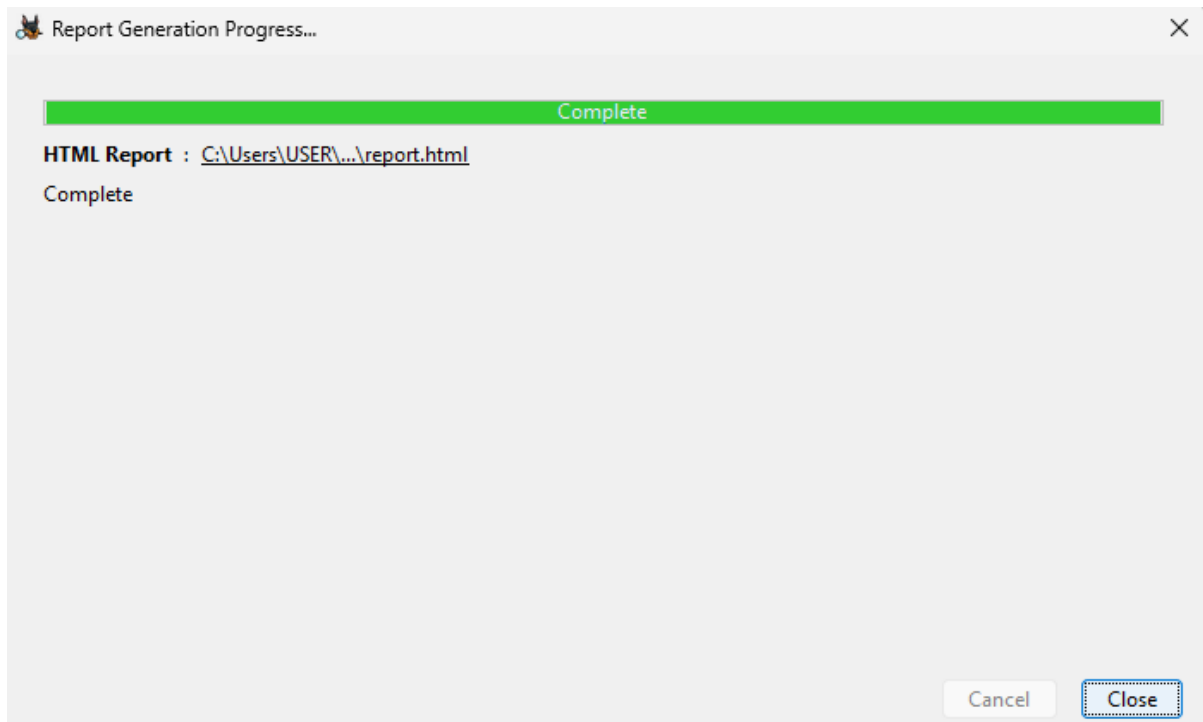
Here is the hidden file found (INCOME.xls)

GENERATION OF REPORT

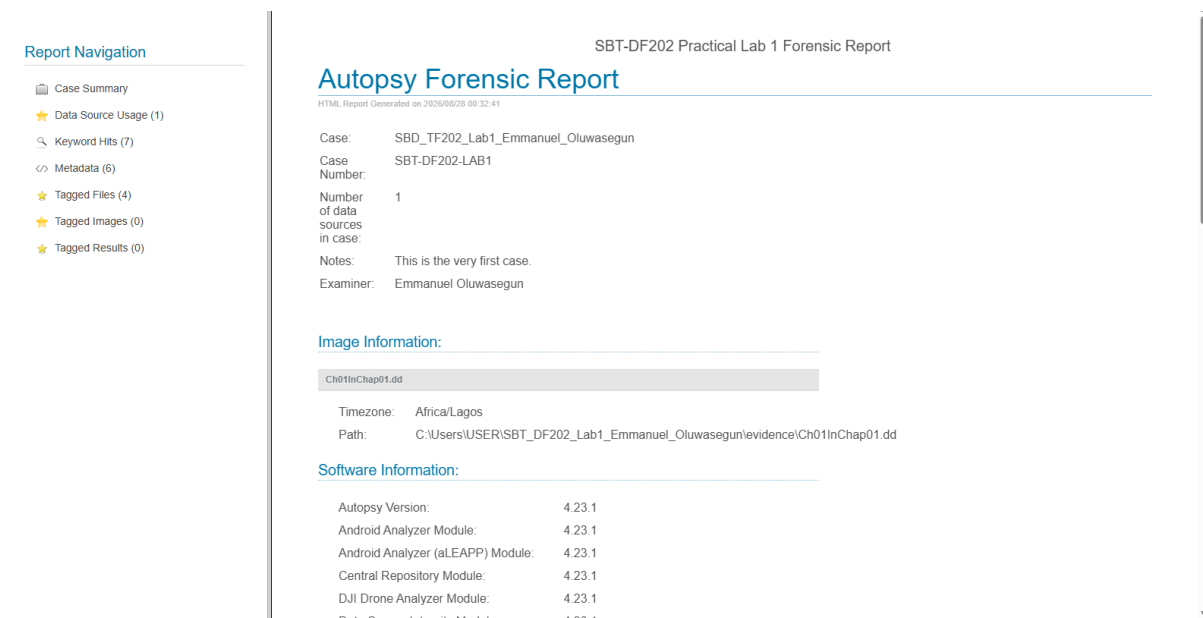
There in the Autopsy the report gets generated also, and below are the steps to generating the report.



This shows the header and footer of the report that would be generated.



This shows the report generation process.



And finally, here's the report.

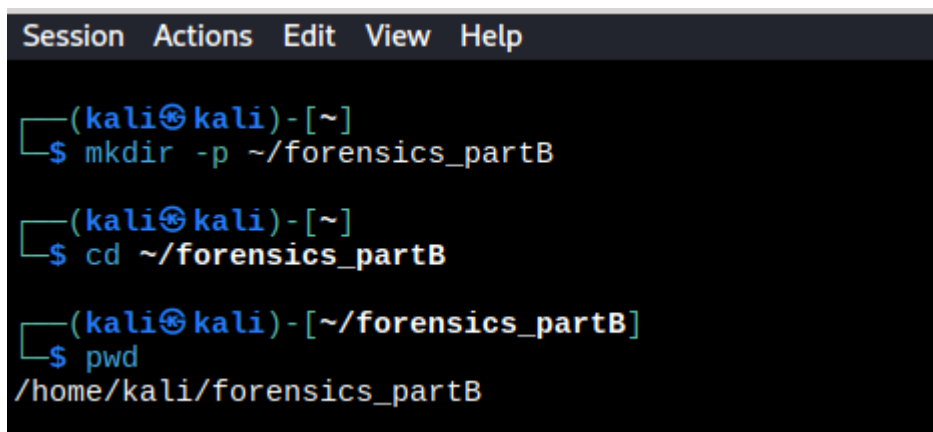
FORENSIC ANALYSIS SUMMARY

Field	Logged evidence/ Recorded details
Case name	SBT_DF202_Lab1_Emmanuel_Oluwasegun
Evidence source	C:\Users\USER\SBT_DF202_Lab1_Emmanuel_Oluwasegun\evidence\Ch01InChap01.dd
Deleted file findings	INCOME.XLS (Identified under deleted unallocated file system entries)
Recovered files	Billing letter.docx, confirmation.txt, income.xls, regrets.docx
Keyword search results	Term: INCOME Total Hits Logged: 7
Reporting evidence	C:\Users\USER\SBT_DF202_Lab1_Emmanuel_Oluwasegun\autopsy_analysis\Report\index.html

SLEUTH KIT ANALYSIS

The sleuth kit analysis is a collection of command line tools alongside a C library which allows one to analyze disk images and recover files from them.

Before starting, I created here in my home folder of kali, a directory which will hold all findings (forensics_partB) as shown in the image below.

A terminal window with a dark background and light blue text. The window has a menu bar at the top with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows three commands being executed: 'mkdir -p ~/forensics_partB', 'cd ~/forensics_partB', and 'pwd'. The output of 'pwd' is '/home/kali/forensics_partB'.

```
Session Actions Edit View Help

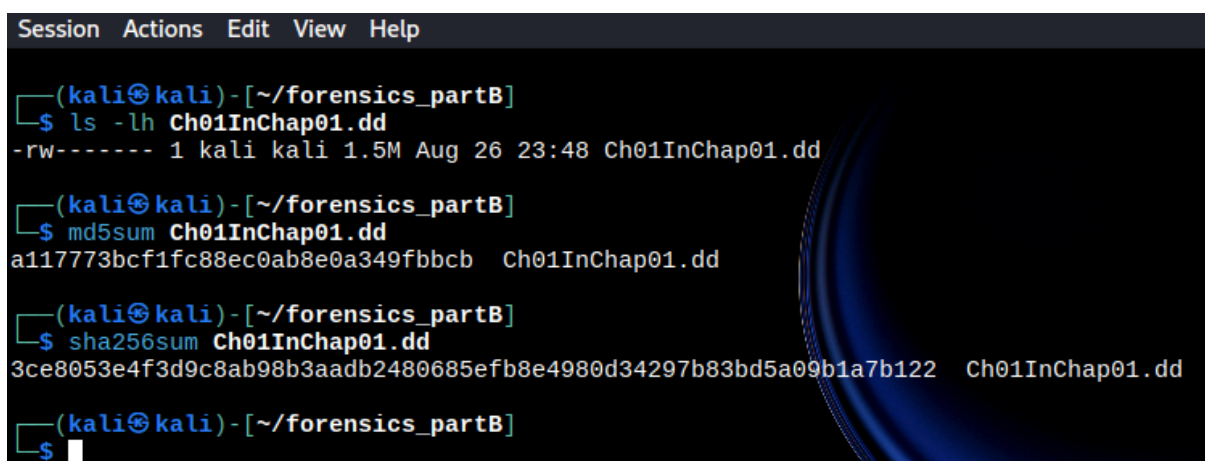
(kali@kali)-[~]
$ mkdir -p ~/forensics_partB

(kali@kali)-[~]
$ cd ~/forensics_partB

(kali@kali)-[~/forensics_partB]
$ pwd
/home/kali/forensics_partB
```

EVIDENCE IMAGE VERIFICATION AND CRYPTOGRAPHIC

HASHING

A terminal window with a dark background and light blue text. The window has a menu bar at the top with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows three commands being executed: 'ls -lh Ch01InChap01.dd', 'md5sum Ch01InChap01.dd', and 'sha256sum Ch01InChap01.dd'. The output of 'ls' shows file details. The output of 'md5sum' shows the MD5 hash. The output of 'sha256sum' shows the SHA256 hash.

```
Session Actions Edit View Help

(kali@kali)-[~/forensics_partB]
$ ls -lh Ch01InChap01.dd
-rw----- 1 kali kali 1.5M Aug 26 23:48 Ch01InChap01.dd

(kali@kali)-[~/forensics_partB]
$ md5sum Ch01InChap01.dd
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd

(kali@kali)-[~/forensics_partB]
$ sha256sum Ch01InChap01.dd
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd

(kali@kali)-[~/forensics_partB]
$
```

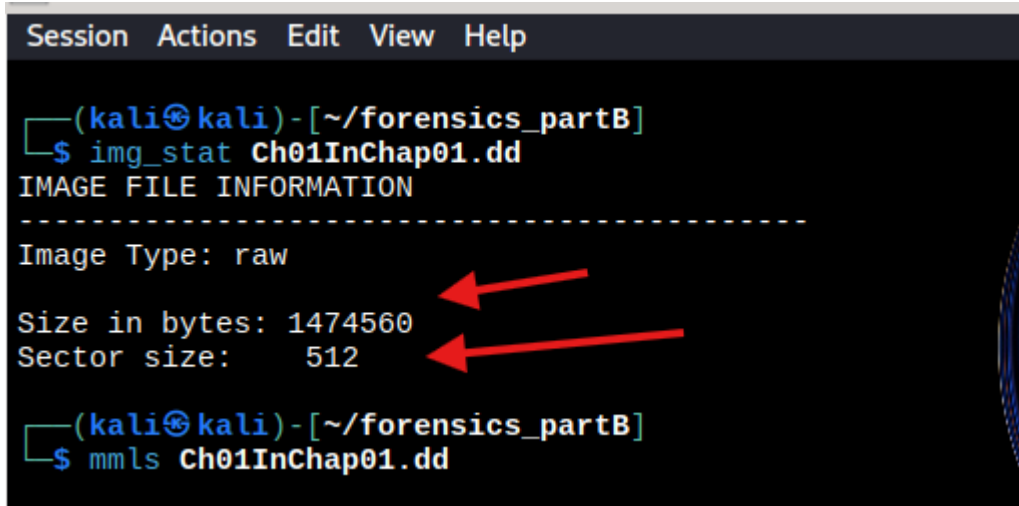
This screenshot shows the initial verification of the evidence image Ch01InChap01.dd.

Running `ls -lh` confirms the file exists in the directory with a size of approximately 1.5MB (1,474,560 bytes).

Cryptographic hashes were generated to establish evidence integrity, producing an MD5 hash of `a117773bcf1fc88ec0ab8e0a349fbbcb` and a SHA-256 hash of `3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122`. These baseline hash values ensure that any future operations on the image can be validated against the original file to guarantee data authenticity throughout the investigation.

DISK IMAGE INFORMATION AND PARTITION TABLE INSPECTION (IMG_STAT & MMLS)

Now to inspect the image, identify the filesystem structure, and recursively list all deleted files, we go through the processes shown below:



```
Session  Actions  Edit  View  Help

(kali㉿kali)-[~/forensics_partB]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size:    512

(kali㉿kali)-[~/forensics_partB]
$ mmls Ch01InChap01.dd
```

This image shows the details of the image format- size in bytes and the sector size. The mmls command shown doesn't give any response which indicates that there is no partition tables.

FILE SYSTEM INFORMATION AND VOLUME LAYOUT ANALYSIS (FSSTAT)

```
(kali㉿kali)-[~/forensics_partB]
└─$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 0300M|

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 9
* FAT 1: 10 - 18
* Data Area: 19 - 2879
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

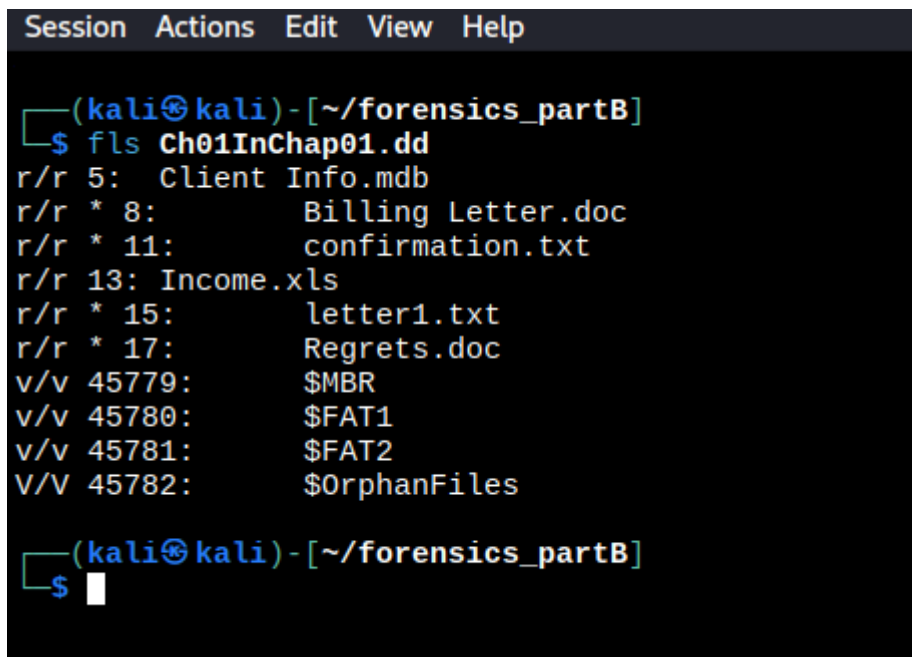
FAT CONTENTS (in sectors)
-----
33-236 (204) -> EOF
285-311 (27) -> EOF
```

The image above shows the details of the file system and we can see the following information from it:

- File System Type: FAT12
- Sector Size: 512 bytes
- Cluster Size: 512 bytes (1 sector per cluster)
- Total Sector Range: 0 to 2879 (2,880 total sectors)
- Boot Sector Location: Sector 0

- File Allocation Tables (FAT):
- FAT 0: Sectors 1 – 9
- FAT 1: Sectors 10 – 18
- Root Directory Sector Range: Sectors 19 – 32
- Cluster Area Start: Sector 33
- Total Cluster Range: 2 – 2848
- Metadata Range: 2 – 45782 (Root Directory Inode/Metadata ID: 2)

DIRECTORY LISTING AND FILE SYSTEM ENUMERATION (FLS)

A screenshot of a Kali Linux terminal window. The window has a title bar with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a prompt '(kali㉿kali)-[~/forensics_partB]' followed by the command '\$ fls Ch01InChap01.dd'. The output lists various files and their attributes: 'r/r 5: Client Info.mdb', 'r/r * 8: Billing Letter.doc', 'r/r * 11: confirmation.txt', 'r/r 13: Income.xls', 'r/r * 15: letter1.txt', 'r/r * 17: Regrets.doc', 'v/v 45779: \$MBR', 'v/v 45780: \$FAT1', 'v/v 45781: \$FAT2', and 'V/V 45782: \$OrphanFiles'. The prompt is repeated at the bottom of the screenshot.

```
Session Actions Edit View Help

(kali㉿kali)-[~/forensics_partB]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

(kali㉿kali)-[~/forensics_partB]
$
```

This screenshot shows the output from running `fls Ch01InChap01.dd` to list the entries within the file system's root directory.

The tool maps out active files, deleted records, and virtual system metadata addresses.

Active files include `Client Info.mdb` (entry 5) and `Income.xls` (entry 13).

Four deleted files are clearly marked with an asterisk (*):

`Billing Letter.doc` (entry 8), `confirmation.txt` (entry 11), `letter1.txt` (entry 15), and `Regrets.doc` (entry 17).

Additionally, virtual addresses are assigned to key structural components like `$MBR`, `$FAT1`, `$FAT2`, and `$OrphanFiles`.

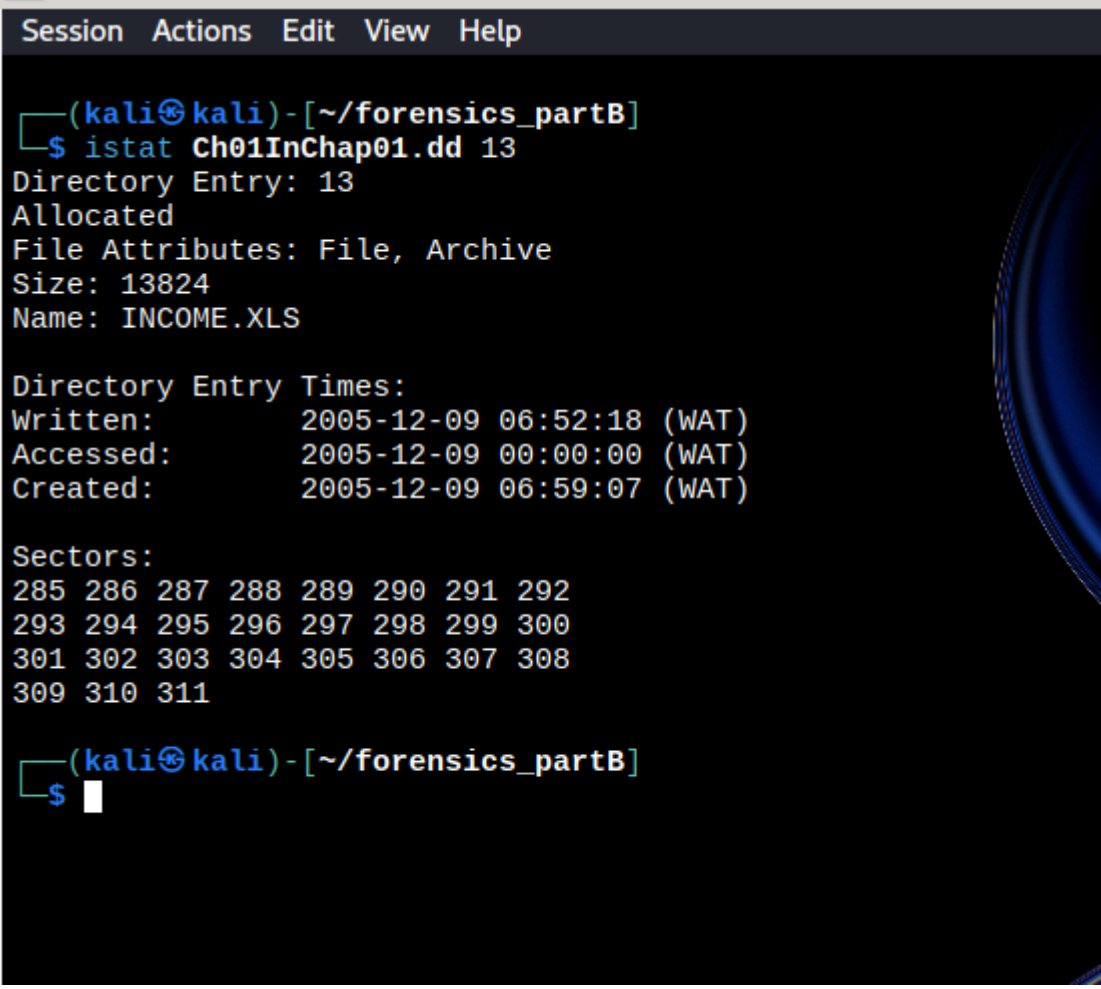
FILE EXTRACTION AND BINARY STREAM INSPECTION (ICAT)

[illegible]

This screenshot demonstrates extracting `income.xls` directly from metadata address 13 using `icat Ch01InChap01.dd 13 > income.xls`, followed by printing its contents using `cat income.xls`. Because an Excel file is a proprietary binary format rather than plain text, viewing it with `cat` dumps raw binary data across the terminal screen alongside embedded text strings.

Despite the squeezed characters, readable strings still show, including author details (Amelia Phillips), staff names (Laura Roper, Randall Watson, Earnest Bell, Frank Haron), financial labels (January Cash Flow, Income, Confirmation, Grand Total), and printer setup strings (HP DeskJet 820C Series Printer).

METADATA ANALYSIS AND DATA UNIT INSPECTION (ISTAT)



```
Session Actions Edit View Help

(kali㉿kali)-[~/forensics_partB]
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written:      2005-12-09 06:52:18 (WAT)
Accessed:     2005-12-09 00:00:00 (WAT)
Created:      2005-12-09 06:59:07 (WAT)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311

(kali㉿kali)-[~/forensics_partB]
$
```

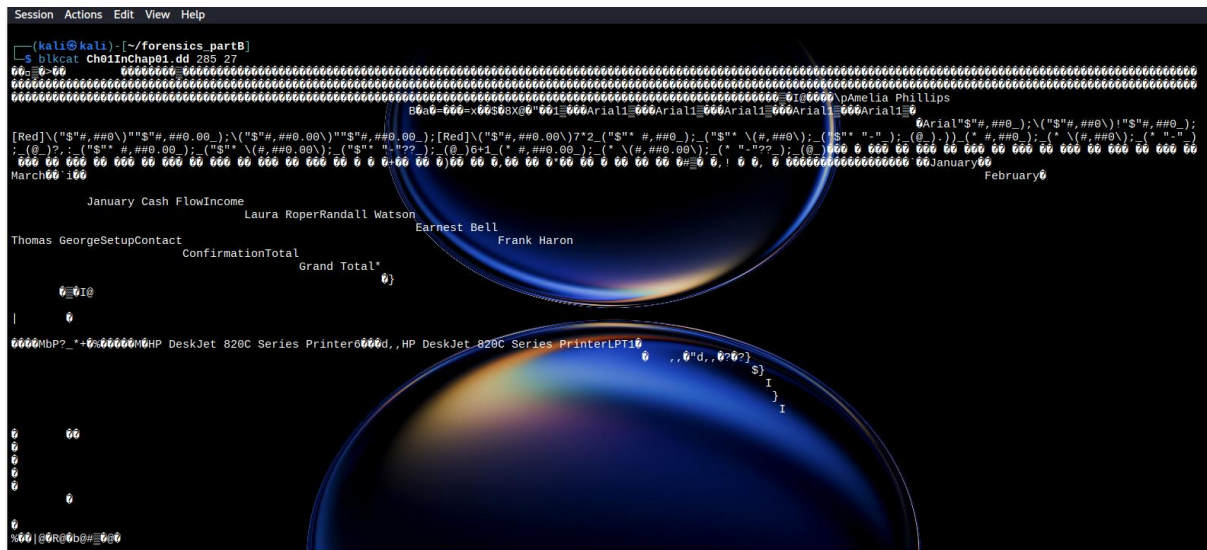
The image above displays the statistics and details about a given meta data structure. Details below:

File Metadata Analysis (istat Ch01InChap01.dd 13)

- Metadata Address (Directory Entry): 13
- Allocation Status: Allocated (An active file entry on the file system)
- File Attributes: Regular File, Archive
- File Size: 13,824 bytes
- Timestamps:
 - Written (Modified): 2005-12-09 06:52:18 (WAT)
 - Accessed: 2005-12-09 00:00:00 (WAT)

- Created: 2005-12-09 06:59:07 (WAT)
- Allocated Data Sectors: Sectors 285 through 311 (27 contiguous 512-byte sectors total: 285–311).

RAW SECTOR DATA EXTRACTION(BLKCAT)



This screenshot shows the terminal output from running blkcat Ch01InChap01.dd 285 27 to extract raw data straight from sectors 285 to 311. Since we dumped the raw sectors to the terminal instead of saving to a file, the binary code shows up as scrambled text mixed with readable strings.

Even in raw form, we can clearly see key file details like author metadata (Amelia Phillips), employee names (Laura Roper, Randall Watson, Earnest Bell, Frank Haron), document titles (January Cash Flow, Income), and printer setup data (HP DeskJet 820C Series Printer).

FILE RECOVERY

```
Session Actions Edit View Help

(kali㉿kali)-[~/forensics_partB]
$ mkdir -p recovered_files
tsk_recover -e Ch01InChap01.dd recovered_files/
Files Recovered: 6

(kali㉿kali)-[~/forensics_partB]
$ ls -la recovered_files/
total 184
drwxrwxr-x 2 kali kali 4096 Aug 28 22:43 .
drwxrwxr-x 3 kali kali 4096 Aug 28 22:43 ..
-rw-rw-r-- 1 kali kali 24064 Aug 28 22:43 'Billing Letter.doc'
-rw-rw-r-- 1 kali kali 104448 Aug 28 22:43 'Client Info.mdb'
-rw-rw-r-- 1 kali kali 227 Aug 28 22:43 confirmation.txt
-rw-rw-r-- 1 kali kali 13824 Aug 28 22:43 Income.xls
-rw-rw-r-- 1 kali kali 121 Aug 28 22:43 letter1.txt
-rw-rw-r-- 1 kali kali 23552 Aug 28 22:43 Regrets.doc

(kali㉿kali)-[~/forensics_partB]
$
```

The image above shows the recovery of the files from the disk image into a local directory that was created (recovered_files) which was successful.

CRYPTOGRAPHIC HASH VERIFICATION AND DATA INTEGRITY ANALYSIS

```
kali@kali: ~/forensics_partB

Session Actions Edit View Help

(kali㉿kali)-[~/forensics_partB]
$ md5sum INCOME_icat.xls INCOME_blkcat.xls recovered_files/Income.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_icat.xls
6a2e65afc5af4fc5f9da2859df134eac INCOME_blkcat.xls
6a2e65afc5af4fc5f9da2859df134eac recovered_files/Income.xls

(kali㉿kali)-[~/forensics_partB]
$ sha256sum INCOME_icat.xls INCOME_blkcat.xls recovered_files/Income.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d INCOME_icat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d INCOME_blkcat.xls
8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d recovered_files/Income.xls

(kali㉿kali)-[~/forensics_partB]
$
```

This screenshot shows the hash checks for our target file, INCOME.XLS, which we pulled using three different Sleuth Kit methods:

metadata extraction (icat), direct sector dumping (blkcat), and automated recovery (tsk_recover).

Running both md5sum and sha256sum gives us the exact same hash values across all three files (6a2e65afc5af4fc5f9da2859df134eac for MD5 and 8d7cd7204d3dae161a8fa879e184865b3bc4a57a4e688abd522a9ff03f62252d for SHA-256).

Because the hashes match perfectly, we have solid proof that the recovered file is 100% complete and completely unchanged.

```
Session Actions Edit View Help
(kali@kali)-[~/forensics_partB]
$ file recovered_files/Income.xls
recovered_files/Income.xls: Composite Document File V2 Document, Little Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Phillips, Last Saved By: Amelia Phillips, Name of Creating Application: Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Saved Time/Date: Fri Dec 9 14:52:18 2005, Security: 0
(kali@kali)-[~/forensics_partB]
$ strings recovered_files/Income.xls | head -n 25
Amelia Phillips
"$#,,##0_);\("$#,,##0\
"$#,,##0_);[Red]\("$#,,##0\
"$#,,##0.00_)\("$#,,##0.00\
"$#,,##0.00_);[Red]\("$#,,##0.00\
_("$# #,##0_);("$# " \($,##0\);_("$#" "-");_(@_)
(* #,##0_);_* \($,##0\);_* "-" _);_(@_)
_("$# #,##0.00_);_("$#" " \($,##0.00\);_("$#" "-"??);_(@_)
_("$# ,##0.00_);_* \($,##0.00\);_* "-"???);_(@_)
January
February
March
January Cash Flow
Income
Laura Roper
Randell Watson
Earnest Bell
Frank Haron
Thomas George
Setup
Contact
Confirmation
Total
Grand Total
MbP?_
(kali@kali)-[~/forensics_partB]
```

Running file confirms the extracted artifact is a valid Composite Document File V2 Document (legacy Microsoft Excel binary format) created on a Windows system. Internal document metadata reveals that Amelia Phillips is listed as both author and last saved user, with an original creation date of November 23, 2002, and a last saved timestamp of December 9, 2005.

- Personnel Names: Laura Roper, Randall Watson, Earnest Bell, Frank Haron, Thomas George, and Amelia Phillips.
- Accounting Labels: January Cash Flow, Income, January, February, March, Total, and Grand Total.

- Worksheet Structure: Setup, Contact, and Confirmation tabs.

This step confirms file structural integrity and verifies that the extracted data matches the target financial records.

EXECUTIVE CONCLUSION

The digital forensic examination of the image file Ch01InChap01.dd successfully recovered and validated the primary target artifact, INCOME.XLS. Using multiple forensic extraction techniques (metadata-based via icat, direct sector dumping via blkcat, and automated volume extraction via tsk_recover), identical output files were extracted from sectors 285–311.

Cryptographic hash verification (md5sum and sha256sum) confirmed 100% data integrity across all extracted copies, proving that the evidence remained uncorrupted throughout analysis.

Metadata and string analysis established file authorship (Amelia Phillips), timestamps (created November 2002, last saved December 2005), and key financial content involving several individuals (Laura Roper, Randall Watson, Earnest Bell, Frank Haron, and Thomas George).

All practical analysis tasks are complete, and evidence chain of custody has been maintained.